

Quiz : 4.1, 4.3

Q 1: find the critical points, domain endpoints, and extreme values (absolute and local) . **(5)**

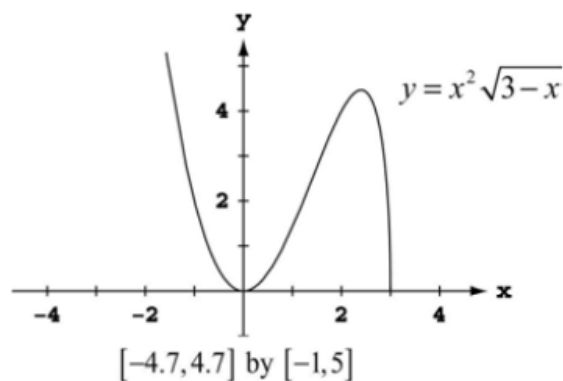
$$y = x^2 \sqrt{3-x}$$

Answer:

$$y' = x^2 \frac{1}{2\sqrt{3-x}} (-1) + 2x\sqrt{3-x} = \frac{-x^2 + (4x)(3-x)}{2\sqrt{3-x}}$$

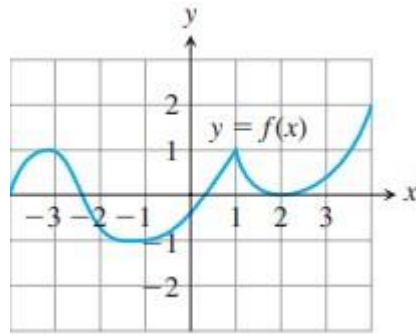
$$= \frac{-5x^2 + 12x}{2\sqrt{3-x}}$$

crit.pt.	derivative	extremum	value
$x = 0$	0	minimum	0
$x = \frac{12}{5}$	0	local max	$\frac{144}{125} 15^{1/2} \approx 4.462$
$x = 3$	undefined	minimum	0



Q2 : a. Find the open intervals on which the function is increasing and decreasing.

b. Identify the function's local and absolute extreme values, if any, saying where they occur... **(5)**



Answer:

- (a) Increasing on $(-4, -3.25)$, $(-1.5, 1)$, and $(2, 4)$, decreasing on $(-3.25, -1.5)$ and $(1, 2)$
(b) Absolute maximum at $(4, 2)$, local maximum at $(-3.25, 1)$ and $(1, 1)$; Absolute minimum at $(-1.5, -1)$, local minimum at $(-4, 0)$ and $(2, 0)$